

LiCT-Net: Lightweight Convolutional Transformer Network for Multiclass Breast Cancer Classification

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Abstract—Classification of multiclass breast cancer through histopathological images is indispensable and poses daunting challenges due to color inconsistencies, high appearance variations, and large inter-class similarities. Regardless of the success of traditional convolutional neural networks (CNNs) and vision transformers (ViTs), they often fail to capture intricate patterns and local contextual information, respectively, while demanding high computational resources and data requirements. To mitigate these issues, this paper proposes a lightweight convolutional transformer network named LiCT-Net for multiclass breast cancer classification. The LiCT-Net introduces a local-global spatially-aware transformer layer and integrates it with a pre-trained FastViT model to effectively capture global and local contextual features, thereby facilitating learning fine-grained and intricate lesion patterns from histopathological images. The LiCT-Net is validated on a benchmark dataset and the experimental results and comparative analysis demonstrate its effectiveness. In specific, it achieves a higher accuracy of 96.16%, 95.62%, 95.25%, and 94.21% on 40 $\times$, 100 $\times$, 200 $\times$, and 400 $\times$ magnifications, respectively.

Index Terms—Breast Cancer, Large Kernel Convolution, LiCT-Net, LSTL, Global Spatial Attention, Local Spatial Attention

I. INTRODUCTION

Breast cancer (BC) is one of the most common cancer types among women across the globe, accounting for nearly 670,000 deaths in 2022 [1]. Early detection and timely treatment of BC is of significant importance to reduce the mortality related to BC. Histopathological imaging is considered the gold standard for confirmation of BC although several imaging modalities, such as ultrasound, mammograms, and MRI, often exist. However, manual assessment of these images is tedious, time-consuming, and error-prone, necessitating the development of automated computer-aided diagnosis (CAD) systems. These systems aim to provide rapid and precise breast cancer identification, alleviating the burden on pathologists. Early efforts on CAD system design relied on manual hand-crafted feature extraction from whole images or segmented patches and subsequent classifier selection [2].

Deep learning methods, especially convolutional neural networks (CNNs), are favored for analyzing histopathologi-

cal images due to their ability to automatically learn hierarchical features [4]. However, the existing CNN approaches often lack end-to-end (E-to-E) learning and fall short in capturing fine-grained details, leading to lower performance. A few attention-based CNN methods [5] have been proposed to address these issues, but there still exist plenty room for potential improvements.

Vision transformers (ViTs), known for their effective modeling of global relationships through multi-head self-attention, have demonstrated superior performance over CNNs in various vision tasks [6]. Limited studies have explored ViT's potential in breast cancer detection due to its challenges, such as high computational demands and limitations in capturing local contextual information [7]. Existing methods, including CNNs and ViTs, primarily address the binary classification task of breast cancer. Yet multiclass classification remains essential and challenging due to structural variations, cancer cell coherence, and color distribution differences. There exists only a limited number of methods proposed to date, and those methods can not learn both local and global contextual features. To this end, we introduce a lightweight convolutional transformer network, LiCT-Net, for multiclass breast cancer classification. It integrates a novel transformer layer with a pre-trained ViT model to effectively capture local and global contextual information, thereby improving classification accuracy. The primary contributions are as follows: (i) we propose LiCT-Net, a lightweight and end-to-end trainable convolutional transformer network, for multiclass breast cancer classification in histopathological images, combining a pre-trained FastViT model with a novel transformer layer, (ii) we introduce a local-global spatially-aware transformer layer (LSTL), mainly comprised of a global spatial attention block (GSAB) and local spatial attention block (LSAB), to explore global and local feature dependencies spatially. The GSAB aims to enlarge the receptive field, which substitutes the well-known self-attention mechanism, thereby improving the model's ability to find complex and diverse lesion patterns in histopathological images, and (iii) we evaluate LiCT-

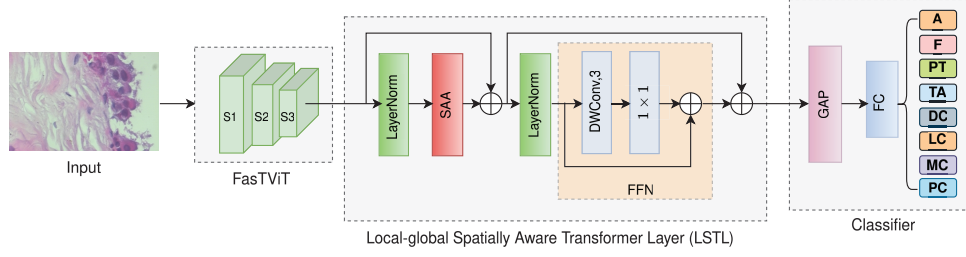


Fig. 1. The overall pipeline of LiCT-Net for multiclass breast cancer classification. The major components include a backbone network (pre-trained FastViT [3]), a LSTL, and a classification layer.

Net using a benchmark dataset and compare its performance against various state-of-the-art CNN and ViT models. Further, ablative experiments are carried out to test the effectiveness of each component of LSTL.

II. PROPOSED METHOD

As shown in Fig. 1, our LiCT-Net architecture primarily consists of three parts: a pre-trained FastViT as a backbone, a loca-global spatially-aware transformer layer (LSTL), and a classification layer for multiclass breast cancer classification. Initially, the histopathological images are passed through the backbone network to extract hierarchical feature maps. We utilized the pre-trained FastViT [3] architecture, a deep convolutional transformer network, as the backbone. It is worth noting that the features have been extracted from the S3-stage of FastViT (S1-S2-S3), resulting in feature maps $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$, where H and W denote the spatial resolution and C indicates the number of channels. Subsequently, $\mathbf{F}$ is processed through the LSTL, which comprises of a global spatial attention block (GSAB) and local spatial attention block (LSAB) followed by a FFN, to extract salient features from key lesion regions, and skip connections for enhanced feature flow. Finally, the refined features are fed into a classification layer, consisting of a global average pooling (GAP) and a dense layer to facilitate the classification of histopathological images into eight breast tumor types.

A. Local-global Spatially Aware Transformer Layer (LSTL)

The proposed LSTL consists of layer normalization (*LayerNorm*), spatially aware attention (SAA), FFN, and skip connections. The major goal of introducing SAA is to comprehensively capture long-range global and local feature dependencies along spatial dimension, thereby enhancing feature representations. The working of LSTL can be mathematically expressed as follows:

$$X' = \mathbf{F} \oplus \text{SAA}(\text{LayerNorm}(\mathbf{F})) \quad (1)$$

$$\text{LSTL}_{out} = X' \oplus \text{FFN}(\text{LayerNorm}(X')) \quad (2)$$

where $\mathbf{F}$ denotes the feature maps generated from backbone. The FFN is formulated as:

$$\text{FFN}_{out} = \text{FFN}_{in} \oplus \text{Conv}_{k=1}(\text{DWConv}_{k=3}(\text{FFN}_{in})) \quad (3)$$

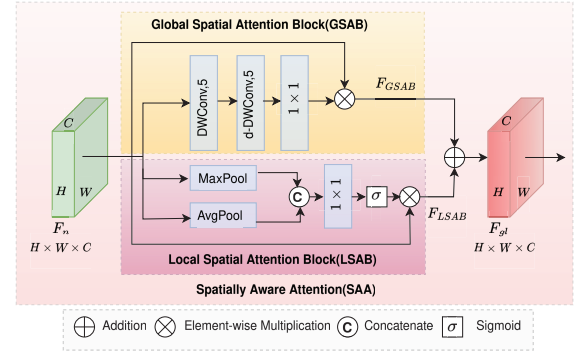


Fig. 2. Illustrative representation of LSTL.

where, $\text{DWConv}(\cdot)$ indicates depthwise convolution with kernel size of k followed by ReLU activation and $\text{Conv}(\cdot)$ denotes the 1×1 pointwise convolution.

1) *Spatially Aware Attention (SAA)*: Lesions in histopathological images exhibit distinct patterns with random spatial distributions. Therefore, extracting these patterns and their relationships is imperative for accurate classification. Self-attention has been a key component in ViT models due to its capability of modeling extensive long-range feature interactions. However, it relies on standard convolution with a fixed kernel size, capturing features within a limited receptive field. Further, it leads to high computational overhead and can capture only global feature dependencies in spatial dimension. To address these issues, we introduce a spatially aware attention (SAA) as depicted in Fig. 2, which comprises a global spatial attention block (GSAB) and a local spatial attention block (LSAB) connected in parallel. The GSAB is inspired by large kernel convolutions [8] and is adopted to capture global contextual information by enlarging the receptive fields, whereas LSAB captures local feature relationships while ignoring irrelevant ones. The output attention feature map F_{gl} is computed as:

$$F_{gl} = F_{GSAB} \oplus F_{LSAB} \quad (4)$$

where, F_{GSAB} and F_{LSAB} indicate the attention feature map

TABLE I
COMPARISON WITH STATE-OF-THE-ART CNN AND ViT MODELS FOR MULTICLASS BREAST CANCER CLASSIFICATION

Model		Magnification Factor												#Params (M)	FLOPs (G)
		40×			100×			200×			400×				
		A_{cc}	P_{re}	R_{ec}	A_{cc}	P_{re}	R_{ec}	A_{cc}	P_{re}	R_{ec}	A_{cc}	P_{re}	R_{ec}		
Pure CNN	EffNet-B6 [9]	92.0	93.0	92.0	92.3	92.6	92.3	88.1	88.5	88.1	74.2	77.9	74.2	43.0	19.0
	RegNetY-16 [10]	92.7	92.7	92.7	92	92.1	92.0	90.9	91.3	90.9	92.3	92.2	92.3	84.0	16.0
	NFNet-F0 [11]	86.6	88.3	86.6	83.5	84.8	83.5	84.1	85.0	84.1	83.7	85.0	83.7	71.4	16.5
	ConvNeXt-T [12]	90.7	90.7	90.7	90.7	90.9	90.7	86.8	87.3	86.8	87.7	88.0	87.7	28.5	8.9
	ConvNeXt-S [12]	90.2	92.3	92.0	90.2	90.8	90.2	86.1	86.7	86.1	88.1	88.3	88.1	50.2	17.4
Pure ViT	ViT-T/32 [6]	92.0	92.3	92.0	91.8	92.1	91.8	88.6	88.7	88.6	88.6	88.6	88.6	5.7	1.3
	CaiT-T [13]	93.5	93.6	93.5	94.1	94.1	94.1	92.1	92.2	92.1	91.2	90.9	91.2	17.3	3.7
	Swin-T [14]	94.2	94.4	94.2	94.4	94.4	94.4	93.1	93.4	93.1	93.2	93.1	93.2	28.3	4.6
	Focal-T [15]	92.3	92.6	92.3	93.3	93.3	93.3	92.7	93.0	92.7	90.8	90.5	90.5	29.1	4.9
	FastViT-T8 [3]	93.8	94.9	94.7	92.16	93.0	92.0	92.88	93.0	93.0	92.7	92.8	92.7	4.0	1.3
LiCT-Net(Ours)		96.16	96.0	96.0	95.62	95.0	95.0	95.25	95.0	95.0	94.21	93.0	93.0	1.1	1.0

generated from GSAB and LSAB, respectively, which are computed as

$$F'_n = \text{Conv}_{k=1}(\text{DWConv}_{k=5}(\text{DWConv}_{k=5}(F_n))) \quad (5)$$

$$F_{\text{GSAB}} = F_n \otimes F'_n \quad (6)$$

where, $\text{DWConv}_{k=5}(\cdot)$ is the depthwise dilated convolution with kernel size of k and dilation rate of 3. The F_{LSAB} is formulated as follows:

$$F''_n = \text{Conv}_{k=1}(\text{Concat}(\text{APool}(F_n), (\text{MPool}(F_n)))) \quad (7)$$

$$F_{\text{LSAB}} = F_n \otimes \sigma(F''_n) \quad (8)$$

where, $\sigma(\cdot)$ is the sigmoid activation and $\otimes$ is the element-wise multiplication. In a nutshell, the SAA offers several benefits: (i) it captures the diversity in attention maps, and (ii) it comprehensively explores local and global feature dependencies while reducing computational overhead.

III. EXPERIMENTS AND RESULTS

A. Dataset and Experimental Setup

To evaluate our LiCT-Net, we used the BreakHis dataset [2], consisting of histopathological images from 82 patients at four magnifications (40×, 100×, 200×, 400×). The dataset contains 7909 histopathological images categorized into adenosis (A), fibroadenoma (F), phyllodes tumor (PT), tubular adenoma (TA), ductal carcinoma (DC), lobular carcinoma (LC), mucinous carcinoma (MC), and papillary carcinoma (PC). The detailed distribution of patients and images corresponding to each breast tumor can be found in [2]. All input images were resized to 256×256 pixels. We used a 60:10:30 train-validation-test split and applied several affine transformations like rotation, rescaling, flips, zooming, and shifts for data augmentation. Experiments were conducted on an NVIDIA Tesla V-100 GPU with 128 GB RAM and models were implemented using Keras and TensorFlow. All models were optimized using Adam optimizer with learning rate as 0.0001 and trained for 100 epochs. The performance metrics used are accuracy (A_{cc}), precision (P_{re}), recall (R_{ec}) and area under the receiver operating characteristics curve (AUC).

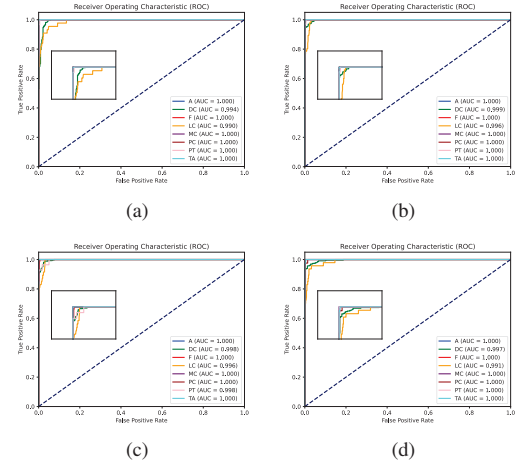


Fig. 3. The ROC curve of LiCT-Net at (a) 40× (b) 100× (c) 200× (d) 400× magnification.

B. Results

1) Comparison with State-of-the-art CNN and ViT Models: The proposed LiCT-Net was compared with various state-of-the-art CNN and ViT models, initially trained on a large-scale image dataset and then fine-tuned on the BreakHis dataset. The CNN models included EffNet [9], EffNetV2 [21], RegNetY-16 [10], NFNet-F0 [11], and ConvNeXt [12], as shown in Table I. Additionally, we also derived a comparative analysis with state-of-the-art ViT models, including ViT [6], CaiT-T [13], Swin-T [14], Focal-T [15], and FastViT-T8 [3]. It can be seen that our LiCT-Net outperformed contemporary CNN and ViT-based models across all magnifications. Notably, it achieved the highest classification accuracy of 96.16%, 95.62%, 95.25%, and 94.21% for 40×, 100×, 200×, and 400×, respectively. Additionally, the class-wise ROC curves with their corresponding AUC values, obtained by LiCT-Net at four different magnifications, as shown in Fig. 3 further validate the efficacy of the proposed method. Further, we compared the computational complexity of our proposed model with others

TABLE II
COMPARISON WITH EXISTING STATE-OF-THE-ART MODELS FOR BREAST CANCER DETECTION, ‘–’ INDICATES NOT REPORTED RESULTS.

Model	Magnification Factor											
	40×			100×			200×			400×		
	A_{cc}	P_{re}	R_{ec}	A_{cc}	P_{re}	R_{ec}	A_{cc}	P_{re}	R_{ec}	A_{cc}	P_{re}	R_{ec}
CSDCNN [16]	92.8	–	–	93.9	–	–	93.7	–	–	92.9	–	–
CNN-SVM [17]	83.79	84.27	87.39	84.48	84.29	84.48	80.83	81.85	80.83	81.03	80.84	81.03
SE-ResNet [18]	94.4	95.25	95.55	94.5	94.51	94.64	93.7	90.71	92.24	92.9	90.74	91.09
DRDA-Net [19]	91.78	91.0	91.3	92.53	92.3	92.3	91.39	91.0	91.7	90.47	90.5	90.5
VGGIN-Net [20]	93.62	93.5	93.5	95.21	95.0	95.0	92.83	92.0	92.1	91.59	91.8	91.7
RDTNet [7]	95.23	95.0	95.0	94.04	94.0	94.0	94.04	95.0	94.0	91.75	92.0	92.0
LiCT-Net(Ours)	96.16	96.0	96.0	95.62	95.0	95.0	95.25	95.0	95.0	94.21	93.0	93.0

TABLE III
RESULTS OF ABLATION STUDY ON LSTL COMPONENTS.

GSAB	LSAB	Magnification Factor											
		40×			100×			200×			400×		
		A_{cc}	P_{re}	R_{ec}	A_{cc}	P_{re}	R_{ec}	A_{cc}	P_{re}	R_{ec}	A_{cc}	P_{re}	R_{ec}
✓	✓	96.16	96.0	96.0	95.62	95.0	95.0	95.25	95.0	95.0	94.21	93.0	93.0
✓	×	94.98	95.0	95.0	93.76	94.0	94.0	92.05	92.0	92.0	91.48	91.0	91.0
×	✓	93.17	93.0	93.0	92.37	92.0	92.0	94.04	94.0	94.0	93.4	93.0	93.0
×	×	93.8	94.9	94.7	92.16	93.0	92.0	92.88	93.0	93.0	92.7	92.8	92.7

as shown in Table I, and it is evident that LiCT-Net is the lighter and faster among all the compared architectures.

2) *Comparison with Existing Breast Cancer Classification Methods:* Table II shows a comparative analysis with existing multiclass breast cancer classification methods on BreakHis dataset. It is evident that our approach achieved superior performance than the competent methods at all magnifications. The methods in [16], [17] employed pre-trained CNNs but lacked E-to-E learning capabilities. The E-to-E trainable approaches [18], [20] and the attention-based method in [19] exhibited comparatively lower performance. On the other hand, the ViT-based method [7] achieved competitive performance. The higher performance of LiCT-Net is majorly attributed to the integration of LSTL for learning intricate lesion patterns in histopathological images. It is noteworthy that the reported performance of existing approaches has been taken directly from their papers, except for [7], [20], and [19] which were validated using current experimental settings.

3) *Ablation Study:* To assess the impact of each component of LSTL, including GSAB and LSAB, we performed a series of ablation experiments, with the results presented in Table III. The results clearly show that excluding GSAB from LSTL reduces performance by 1.68%, 1.86%, 3.2%, and 2.73% for 40×, 100×, 200×, and 400×, respectively due to the loss of local contextual information essential for recognizing complex lesion patterns. Similarly, removing LSAB leads to a decreased performance because of the lack of global feature dependencies. Overall, we achieved a higher performance when incorporating both modules into LSTL.

IV. CONCLUSION

This paper introduced LiCT-Net, a lightweight convolutional transformer network, specifically designed for mul-

ticlass classification of breast cancer via histopathological images. The LiCT-Net integrated a novel LSTL with a backbone ViT model to thoroughly capture both global and local contextual details using GSAB and LSAB, capturing lesion relationships within the images and thereby enhancing feature representations. The LiCT-Net supports E-to-E training and is computationally efficient. Experimental results and comparative analyses on a benchmark dataset demonstrated that LiCT-Net is superior than existing state-of-the-art CNN and ViT models. Further, the ablation experiments verified the effectiveness of our proposed LSTL. In future, the potency of LiCT-Net can be tested on other medical image classification tasks.

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